

Recording Activity: Manual Voltage Mapping

Name: _____ Date: _____

Predict

What do you predict will happen to the measured voltage as the control moves from one end to the other?

Measure and Map

Record the minimum, maximum, and at least six additional readings. Use the lesson lookup table to fill in the last two columns. Your completed table should include all eight decimal levels.

Control position	Measured voltage	Decimal level (0-7)	3-bit value
Minimum			
Reading 2			
Reading 3			
Reading 4			
Reading 5			
Reading 6			
Reading 7			
Maximum			

Compare and Explain

1. Find two nearby voltage readings that have the same decimal level. Why can both readings share one level?
2. How many different decimal levels are in this activity? How many 3-bit values?
3. In your own words, explain how the lookup table turns a smoothly changing voltage into digital choices.